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Rotation of particle spin in a storage ring with a polarized beam and measurement of the particle EDM, tensor polarizability and elastic zero-angle scattering amplitude

V G Baryshevsky

Research Institute for Nuclear Problems, Belarus State University, 11 Bobruyskaya Str.,
Minsk 220050, Belarus

E-mail: bar@inp.minsk.by

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Abstract

Equations of particle spin motion in a storage ring are obtained considering contributions from the tensor electric and magnetic polarizabilities of the particle along with contributions from the spin rotation and birefringence effects in a polarized internal target. The study of the spin rotation and birefringence effects for a particle in a high-energy storage ring provides for measuring both the spin-dependent real part of the coherent elastic zero-angle scattering amplitude and tensor electric (magnetic) polarizabilities.

1. Introduction

The investigation of spin-dependent interactions of elementary particles at high energies is a very important part of programs for scientific research on storage rings [1–3]. Such studies are carried out with polarized particle beams and polarized targets. The dependence of scattering cross-sections on the particle spin is the subject of many studies. Experiments for measuring the spin-dependent part of the forward scattering amplitude are in preparation now [2, 3].

It is well known in experimental particle physics how to measure a spin-dependent cross-section. However, measurement of the spin-dependent part of the forward scattering amplitude remains a complicated challenge.

An unambiguous and direct method for measuring the real part of the spin-dependent forward scattering amplitude for high-energy particles was proposed in [4–10]. In particular, the real part of the spin-dependent forward scattering amplitude for protons (deuterons, antiprotons) passing through a polarized nuclear target can be found by means of measuring the particle spin rotation. The deuteron birefringence effect (i.e. the phenomenon of deuteron spin rotation and oscillation) provides for measuring the real part of the spin-dependent forward scattering amplitude for deuterons passing through a nonpolarized target.

A similar method for measuring the real part of the spin-dependent forward scattering amplitude was used in [12–14] for thermal neutrons passing through polarized targets. It was based on the phenomenon of nuclear precession of the neutron spin in the nuclear pseudomagnetic field of the polarized target predicted in [11] and experimentally observed in [12–14].

According to [5, 6], the frequency of the spin rotation Ω_{nuc} for a particle passing through a target with polarized nuclei can be expressed as

$$\Omega_{\text{nuc}} = \frac{2\pi\hbar}{m\gamma} N P_t \Delta f, \quad (1)$$

where m is the mass of the particle, γ is the Lorentz factor, N is the number of nuclei (scatterers) in cm^3 , P_t is the degree of polarization of target nuclei, Δf is the difference between the amplitudes of forward scattering for particles with oppositely directed spins.

For a particle (proton, deuteron) with energy within the range from hundreds MeV to several GeV, the difference $\Delta f \sim 10^{-13} \text{ cm}$. Therefore, Ω_{nuc} for the polarized solid target ($N = 10^{22}$, $P_t = 0.8$) can be evaluated to be $\Omega_{\text{nuc}} \approx 5 \times 10^6 \text{ cm}^{-1}$. Then, the particle with energy from hundreds MeV to several GeV passing the distance L in the polarized solid target acquires spin rotation by the angle

$$\vartheta = \Omega_{\text{nuc}} \frac{L}{v} = 2 \times 10^{-4} \cdot L, \quad (2)$$

where v is the particle velocity.

Therefore, in the polarized solid target of length 20–50 cm spin rotates by the angle $\vartheta \approx 4 \times 10^{-3} \text{--} 10^{-2} \text{ rad}$ [5, 6]. Deuteron birefringence in a nonpolarized target causes spin rotation by a similar angle [7, 8].

The density of the internal gas target in the storage ring is significantly lower than the density of the solid target. As a result, the frequency Ω_{nuc} of particle spin rotation in the polarized gas target (for example, ANKE, COSY) appears smaller $\Omega_{\text{nuc}} \approx 10^{-4} \text{--} 10^{-5}$ [3, 10]. At first glance this value is too small to be measurable. But particles in the storage ring pass through the target many times accumulating the effect. For example, the spin of a particle circulating in the storage ring during the time $t \approx 10^2 \text{ s}$ rotates by the angle $\vartheta = \Omega_{\text{nuc}} t \approx 10^{-3} \text{--} 10^{-2}$. Such values of the rotation angle can be measured by modern techniques for beam polarization measurements (see, for example, measurements of deuteron beam parameters in COSY [15]).

Now the principal method for measuring the deuteron beam polarization is based on the right–left asymmetry in deuteron–nucleus elastic scattering [16]. The requirements for a polarimeter for the deuteron EDM search using a storage ring were analyzed in detail [17], and it was shown that a polarimeter with sensitivity, which offers the possibility of such a search, can be developed. Application of this method has allowed us to take the first step in the experimental study of the above-discussed new physical phenomena [18] and demonstrate that small values of tensor polarization can be measured. In [18] the phenomenon of deuteron spin dichroism was experimentally observed for the first time for 20 MeV deuterons passing through the nonpolarized carbon target. Moreover, according to [2, 3, 15] COSY is now embarking on a program of investigations with polarized deuteron beams and polarized deuterium storage cell targets at the ANKE magnetic spectrometer, placed at the internal target station of the COSY ring.

It should be mentioned that the behavior of the particle spin in a storage ring gets complicated by the influence of electromagnetic fields which are present in the storage ring. They cause spin rotation with the frequency about Megahertz. Moreover, in the real accelerator, due to the unavoidable tolerance of manufacture and alignment, the actual closed orbit is

perturbed slightly from the design orbit [19–24]. As a result, reasonable doubts arise as to whether it is possible to measure new physical effects which are very interesting but revealing themselves by precession with frequencies, which are much smaller than the frequency of particle spin precession in electromagnetic fields in the storage ring.

In this connection I would like to remind here that the experiment to measure the deuteron electric dipole moment is in active preparation now [17, 25, 26]. According to the plan of this experiment, the limits for the deuteron EDM are expected to be obtained at the level 10^{-29} e cm by measuring additional deuteron spin precession in the storage ring caused by EDM interaction with the electromagnetic fields. Contribution from the EDM to the frequency of spin precession can be estimated to be $\Omega_d \approx 10^{-7}–10^{-9} \text{ s}^{-1}$. This value is noticeably smaller than that expected in the proposed experiment for the observation of the nuclear spin precession and birefringence effects ($\Omega_{\text{nuc}} \sim 10^{-4}–10^{-5}$). It is important that the method [17, 25, 26], which is supposed for the study of deuteron spin rotation due to the EDM, can entirely be applied for the observation of the deuteron birefringence effect (i.e. deuteron spin rotation and dichroism in a nonpolarized target). Moreover, as mentioned in [10], the study of the deuteron birefringence effect in experiments for the EDM search is essential to distinguish contributions from the deuteron birefringence effect and deuteron EDM to spin rotation.

Thus, the modern experimental technique allows us to observe the nuclear spin precession and birefringence effects and, therefore, to measure the spin-dependent part of the amplitude of elastic coherent scattering at zero angle. It should also be mentioned that, according to [10, 27–29], in both the EDM experiments and studies of deuteron spin rotation in storage rings one should take into account deuteron electric and magnetic polarizabilities. In the conditions of the EDM experiment [17, 26, 25] these polarizabilities can mix up the effect from EDM, because they cause spin precession with frequency $\Omega \approx 10^{-5}–10^{-7} \text{ s}^{-1}$, which exceeds the expected value of the effect from EDM $\Omega_d \approx 10^{-7}–10^{-9} \text{ s}^{-1}$. In other words, in the EDM experiments the effects, which are discussed in the present paper, will be *a fortiori* observed. This possibility enriches the experiment dedicated for EDM search. The study of the above effects has its own importance because they allow us to measure the electric and magnetic polarizability of a deuteron as well as the spin-dependent part of the amplitude of elastic coherent forward scattering of a deuteron by a nucleus. But, they should be thoroughly scrutinized when analyzing the results of the EDM experiment, because contributions from these effects can hamper correct experiment interpretation.

The theory which describes the motion of the particle spin in electromagnetic fields in a storage ring has been developed in many papers [19–24]. According to [19–24], the basic equation which describes particle spin motion in an electromagnetic field is the Thomas–Bargmann–Michel–Telegdi (T-BMT) equation. Improvement of the T-BMT equation, allowing us to consider the possible presence of the particle EDM, was made in [24, 30]. Supplementary terms [27–29] added to the T-BMT equation also make it describe several new effects.

The study of these effects allows us to measure the spin-dependent part of the zero-angle coherent elastic scattering amplitude as well as the static electric and magnetic polarizabilities of particles.

In the present paper equations describing the spin motion of high-energy particles moving in a storage ring are obtained and additional terms in these equations are analyzed.

This paper is organized as follows. In section 2 the interaction of a deuteron with electric and magnetic fields caused by electric and magnetic polarizabilities is discussed. Section 3 considers the relation between the refraction index and the effective potential energy of a high-energy particle colliding with a bunch of high-energy particles moving in a storage ring. The phenomenon of spin rotation is discussed in section 4 for a proton (deuteron, antiproton)

in a storage ring with a polarized target inside. In section 5 the equations describing the spin motion of a particle in a storage ring are considered. Contributions from the EDM and tensor electric and magnetic polarizabilities to particle spin rotation are considered by the example of deuteron. It is shown that the deuteron polarizabilities can be measured in the deuteron EDM search experiment. Also it is shown that deuteron polarizabilities can imitate the effect from EDM. It is demonstrated that the measurement of the deuteron spin polarization tensor allows us to find the contribution from the deuteron EDM to spin rotation, which does not depend on polarizabilities. This is in contrast to the measurement of the vertical spin component. The equations for particle spin motion in a storage ring with an internal target are considered in section 6. The angle of particle spin rotation in the target is evaluated. It is shown that the considered effects can be observed with modern techniques and used for measuring the spin-dependent part of the amplitude of elastic coherent forward scattering.

2. Particle spin rotation in an electromagnetic field in a storage ring

Let us consider now a particle with spin S moving in the electromagnetic field of a storage ring. The term ‘particle spin’ here means the expectation value of the quantum mechanical spin operator $\hat{S}$ (for more details, see [24]); hereinafter a symbol with a ‘hat’ means a quantum mechanical operator.

First of all, a detailed analysis of such motion shows that the orbital motion of the particle does not depend on the motion of spin and is described by the Lorentz equation, while spin motion depends crucially on orbit motion [20–23]. Spin motion is described by the Thomas–Bargmann–Michel–Telegdi equation (T-BMT) [19, 24] as follows:

$$\frac{d\vec{S}}{dt} = [\vec{S} \times \vec{\Omega}], \quad (3)$$

$$\vec{\Omega} = \frac{e}{mc} \left[\left(a + \frac{1}{\gamma} \right) \vec{B} - a \frac{\gamma}{\gamma + 1} (\vec{\beta} \cdot \vec{B}) \vec{\beta} - \left(a + \frac{1}{\gamma + 1} \right) \vec{\beta} \times \vec{E} \right] \quad (4)$$

where $\vec{S}$ is the spin vector, t is the time in the laboratory frame, m is the mass of the particle, e is its charge, γ is the Lorentz factor, $\vec{\beta} = \vec{v}/c$, where $\vec{v}$ denotes the particle velocity, $\vec{E}$ and $\vec{B}$ are the electric and magnetic fields at the point of particle location in the laboratory frame, $a = (g - 2)/2$ and g is the gyromagnetic ratio. The parameter a for protons is $a = 1.79$ and for deuterons $a = -0.14$. As a result, for the typical value of the field in a storage ring ($B \approx 10^2 - 10^4$ gauss), the spin rotation frequency is $\Omega \approx 10^5 - 10^7$ s⁻¹.

Equation (4) was derived by Thomas [31] (also by Frenkel [32]) and, in a more general form, by Bargmann *et al* [33].

The T-BMT equation describes the motion of the spin in the rest frame of the particle, wherein the spin is described by the three component vector $\vec{S}$ [24]. In practice, the T-BMT equation works well for the description of spin precession in the external electric and magnetic fields encountered in typical modern day accelerators. The study of the T-BMT equation allows us to determine the major peculiarities of spin motion in an external electromagnetic field. However, it should be taken into account that particles in an accelerator have energy spread and move over different orbits. This necessitates one to average the spin-dependent parameters of the particle over the particle beam phase space. This is why one must always bear in mind the distinction between the beam polarization and the spin vector $\vec{S}$. Complete description of particle spin motion can be done applying density matrices (for more details, see [20, 21, 24]).

If a particle possesses an intrinsic electric dipole moment, then the additional term, describing spin rotation induced by EDM, should be added to (3) (see [30]):

$$\frac{d\vec{S}_{\text{edm}}}{dt} = \frac{d}{\hbar} [\vec{S} \times (\vec{\beta} \times \vec{B} + \vec{E})], \quad (5)$$

where d is the electric dipole moment of the particle.

As a result, the motion of the particle spin due to the magnetic and electric dipole momenta can be described by the following equation:

$$\begin{aligned} \frac{d\vec{S}}{dt} = \frac{e}{mc} \left[\vec{S} \times \left\{ \left(a + \frac{1}{\gamma} \right) \vec{B} - a \frac{\gamma}{\gamma + 1} (\vec{\beta} \cdot \vec{B}) \vec{\beta} \right. \right. \\ \left. \left. - \left(a + \frac{1}{\gamma + 1} \right) \vec{\beta} \times \vec{E} \right\} \right] + d[\vec{S} \times (c\vec{\beta} \times \vec{B} + \vec{E})] \end{aligned} \quad (6)$$

The typical frequency ω_d of precession caused by the interaction of the particle EDM with the electromagnetic field is many orders less than Ω . In the conditions of the deuteron EDM experiment [25, 26], the EDM value is expected to be measured at the level $d \approx 10^{-29}$ e cm. Therefore, the expected precession frequency can be evaluated as $\omega_d \approx 10^{-8} \times 10^{-9} \text{ s}^{-1}$. Nevertheless, the change of deuteron spin direction in the storage ring can be measured even for such a small ω_d (see [17, 25, 26]).

The motion of the particle spin for particles with spin $S \geq 1$ in the storage ring is strongly influenced by several additional interactions [10, 27–29]. In particular, a particle with spin $S \geq 1$ possesses electric and magnetic polarizabilities. Particle interaction with the electromagnetic field in the storage ring due to particle electric polarizability can be described as follows:

$$\hat{V}_{\vec{E}} = -\frac{1}{2} \hat{\alpha}_{ik} (E_{\text{eff}})_i (E_{\text{eff}})_k, \quad (7)$$

where $\hat{V}_{\vec{E}}$ is the energy of interaction, $\vec{E}$ and $\vec{B}$ are supposed to be orthogonal to the particle momentum (this case is realized in the storage ring), $\hat{\alpha}_{ik}$ is the electric polarizability tensor of the particle, $\vec{E}_{\text{eff}} = (\vec{E} + \vec{\beta} \times \vec{B})$ is the effective electric field at the point of particle location.

Expression (7) can be rewritten as follows:

$$\hat{V}_{\vec{E}} = \alpha_S E_{\text{eff}}^2 - \alpha_T E_{\text{eff}}^2 (\hat{S} \cdot \vec{e}_E)^2, \quad \vec{e}_E = \frac{\vec{E} + \vec{\beta} \times \vec{B}}{|\vec{E} + \vec{\beta} \times \vec{B}|} \quad (8)$$

where $\hat{S}$ is the spin operator, α_S is the scalar electric polarizability and α_T is the tensor electric polarizability of the particle.

The mentioned interaction results in rotation and oscillations of the spin of the particle moving in the electric field [10, 27–29]. The frequency of this rotation ω_{TE} is determined by the tensor electric polarizability α_t . Theoretical evaluations [34] give for the deuteron polarizability $\alpha_t \sim 10^{-40} \text{ cm}^3$. Therefore, for a deuteron in a storage ring the typical energy (in the frequency units) of spin interaction with the electric field due to the tensor electric polarizability is about $\omega_{\text{TE}} \sim 10^{-5} \text{ s}^{-1}$ in the field $E_{\text{eff}} \sim 10^4 \text{ V cm}^{-1}$.

A particle with spin $S \geq 1$ also possesses magnetic polarizability, which is described by the magnetic polarizability tensor $\hat{\beta}_{ik}$ and the interaction of the particle with the magnetic field due to the magnetic polarizability is as follows:

$$\hat{V}_{\vec{B}} = -\frac{1}{2} \hat{\beta}_{ik} (B_{\text{eff}})_i (B_{\text{eff}})_k, \quad (9)$$

where $(B_{\text{eff}})_i$ are the components of the effective magnetic field $\vec{B}_{\text{eff}} = (\vec{B} - \vec{\beta} \times \vec{E})$. The energy of interaction $\hat{V}_{\vec{B}}$ (9) can be expressed as

$$\hat{V}_{\vec{B}} = \beta_S B_{\text{eff}}^2 - \beta_T B_{\text{eff}}^2 (\hat{S} \cdot \vec{e}_B)^2, \quad \vec{e}_B = \frac{\vec{B} - \vec{\beta} \times \vec{E}}{|\vec{B} - \vec{\beta} \times \vec{E}|}, \quad (10)$$

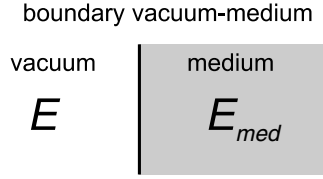


Figure 1. The particle energy in the vacuum E is not equal to that in the medium E_{med} .

where β_S is the scalar magnetic polarizability and β_T is the tensor magnetic polarizability of the particle.

Similarly to the particle's interaction with the electric field, its interaction with the magnetic field arouses rotation and oscillations of the particle spin with frequency ω_t^μ . According to calculations [34], the tensor magnetic polarizability can be evaluated as $\beta_t \sim 2 \times 10^{-40} \text{ cm}^3$. Therefore, this contribution provides rotation with the frequency $\omega_t^\mu \sim 10^{-5} \text{ s}^{-1}$ in the field $B_{eff} \sim 10^4$ gauss.

As is obvious, the frequencies ω_{TE} and ω_t^μ are much higher than the frequency ω_d , which can be expected from deuteron EDM, if it is nonzero. Therefore, the influence of the tensor polarizabilities on particle spin motion in storage rings should be considered with particular attention.

3. The index of refraction and effective potential energy of particles in the medium

Recall now that a storage ring can contain matter inside (gas, jet target). The presence of such a target influences spin motion [10, 27–29]. The following reasoning could clear target influence on spin motion: the close connection between the coherent elastic scattering amplitude at zero angle $f(0)$ and the refraction index of the medium n has been established as a result of numerous studies (see, for example, [35, 36]):

$$n = 1 + \frac{2\pi N}{k^2} f(0) \quad (11)$$

where N is the number of particles per cm^3 and k is the wave number of the particle incident on the target.

Expression (11) is derived in assumption that $n - 1 \ll 1$. If $k \rightarrow 0$, then $(n - 1)$ grows and the refraction index can be expressed as

$$n^2 = 1 + \frac{4\pi N}{k^2} f(0).$$

Let us consider particle refraction on the vacuum–medium boundary (see figure 1). The wave number of the particle in the vacuum is denoted k . The wave number of the particle in the medium is $k' = kn$. As is evident, the particle momentum in the vacuum $p = \hbar k$ is not equal to the particle momentum in the medium. Therefore, the particle energy in the vacuum $E = \sqrt{\hbar^2 k^2 c^2 + m^2 c^4}$ is not equal to the particle energy in the medium $E_{med} = \sqrt{\hbar^2 k^2 n^2 c^2 + m^2 c^4}$.

The energy conservation law immediately necessitates the particle in the medium to have an effective potential energy V_{eff} (see the detailed theory in [36]). This energy can be easily found from the relation

$$E = E_{med} + V_{eff},$$

i.e.,

$$V_{\text{eff}} = E - E_{\text{med}} = -\frac{2\pi\hbar^2}{m\gamma} N f(E, 0) = (2\pi)^3 N T(E), \quad (12)$$

where $T(E)$ is the T-matrix [36] and $f(E, 0) = -(2\pi)^2 \frac{E}{c^2\hbar^2} T(E) = -(2\pi)^2 \frac{m\gamma}{\hbar^2} T(E)$.

Above I have discussed the rest target. But a bunch, which moves in a storage ring, can also be considered as a target. Therefore, expressions (11) and (12) should be generalized for this case. Let E_1 and γ_1 denote the energy and Lorentz factor of the particles of the first beam in the rest frame of the storage ring and E_2 and γ_2 denote those of the particles of the second beam. Bearing in mind that the wave phase in the medium is Lorentz invariant, one can find it in the following way: let the second beam rest in the reference frame, then the refraction index can be expressed in the conventional form (11)

$$n'_1 = 1 + \frac{2\pi N'_2}{k_1'^2} f(E'_1, 0), \quad (13)$$

where $N'_2 = \gamma_2^{-1} N_2$ is the density of bunch 2 in its rest frame and N_2 is the density of the second bunch in the storage ring frame, k'_1 and E'_1 are the wavenumber and energy of particles of the first bunch in the rest frame of bunch 2, respectively. Let L denote the length of bunch 2 in its rest frame, then $L = \gamma_2 l$, where l is the length of this bunch in the storage ring frame.

The interaction of a particle from bunch 1 (particle 1) with the particles of bunch 2 causes a change in the phase of the wave:

$$\phi = k'_1(n'_1 - 1)L = \frac{2\pi N'_2}{k_1'} f(E'_1, 0)L = \frac{2\pi N_2}{k_1} f(E'_1, 0)k'_1 l. \quad (14)$$

It is known [36] that the ratio $\frac{f(E'_1, 0)}{k_1'}$ is invariant, i.e. $\frac{f(E'_1, 0)}{k_1'} = \frac{f(E_1, 0)}{k_1}$, where $f(E_1, 0)$ is the amplitude of the elastic coherent forward scattering of particle 1 by the moving particles of bunch 2 in the rest frame of the storage ring.

As a result,

$$\phi = \frac{2\pi N_2}{k_1} f(E_1, 0) \cdot l = \frac{2\pi N_2}{k_1} f(E_1, 0) \cdot v_{\text{rel}} \cdot t, \quad (15)$$

where v_{rel} is the velocity of the relative motion of particle 1 and bunch 2 (for opposing motion $v_{\text{rel}} = (v_1 + v_2)(1 + \frac{v_1 v_2}{c^2})^{-1}$) and t is the time of interaction of particle 1 with bunch 2 in the rest frame of the storage ring.

The particle with velocity $v_1 = \frac{\hbar k_1 c^2}{E_1}$ passes the distance $z = v_1 \cdot t$ over time t . Note that the distance z differs from the length of bunch 2, because it moves. Expression (15) can be rewritten as

$$\phi = \frac{2\pi N_2}{k_1} f(E_1, 0) \frac{v_{\text{rel}}}{v_1} z = k_1(n_1 - 1)z, \quad (16)$$

where the index of refraction of particle 1 by the beam of moving particles 2 is

$$n_1 = 1 + \frac{2\pi N_2}{k_1^2} \frac{v_{\text{rel}}}{v_1} f(E_1, 0). \quad (17)$$

When $v_2 = 0$, (17) converts to the conventional expression (11).

Thus, the effective potential energy V_{eff} being acquired by particle 1 when it collides with the particles of bunch 2 can be written as follows:

$$\begin{aligned} V_{\text{eff}} &= E_1 - E_{1\text{med}} = E_1 - \sqrt{p_1^2 c^2 n_1^2 + m_1^2 c^4} \\ &= -2\pi\hbar^2 N_2 v_{\text{rel}} \frac{f(E_1, 0)}{p_1} = -2\pi\hbar^2 N_2 v_{\text{rel}} \frac{f(E'_1, 0)}{p'_1}. \end{aligned} \quad (18)$$

Therefore,

$$V_{\text{eff}} = -\frac{2\pi\hbar^2 N_2}{m_1\gamma_1\gamma_2} f(E'_1, 0) = (2\pi)^3 N_2 T(E'_1), \quad (19)$$

where $E'_1 = m_1 c^2 \gamma_1 \gamma_2$ is the energy of particle 1 in the rest frame of bunch 2, p_1 denotes the momentum of particle 1 in the storage ring frame, while p'_1 is the momentum of particle 1 in the rest frame of bunch 2 ($p'_1 = \frac{E'_1 v'_1}{c^2}$) and $v'_1 = v_{\text{rel}}$.

When obtaining (18) it was used $|n_1 - 1| \ll 1$.

Let us consider now a particle with the nonzero spin. In this case the amplitude of the zero-angle scattering depends on the particle spin and, as a consequence, the index of refraction depends on the particle spin and can be written as

$$\hat{n}_1 = 1 + \frac{2\pi N_2}{k_1^2} \frac{v_{\text{rel}}}{v_1} \hat{f}(E_1, 0), \quad (20)$$

where $\hat{f}(E_1, 0) = \text{Tr} \hat{\rho}_J \hat{F}(0)$ is the operator of the forward scattering amplitude which acts in the combined spin space of the particle and scatterer spin, $\hat{\rho}_J$ is the spin density matrix of the scatterers.

According to the above (see (12) and (19)), a particle in matter possesses some effective potential energy V_{eff} . If the amplitude $\hat{f}(0)$ of particle scattering depends on the particle spin, then the effective energy depends on the spin orientation [4–10]:

$$\hat{V}_{\text{eff}} = -\frac{2\pi\hbar^2 N_2}{m_1\gamma_1\gamma_2} \hat{f}(E'_1, 0). \quad (21)$$

For example, the scattering amplitude $\hat{f}(0)$ of a particle with spin $S = 1$ (for example, deuteron) even in an unpolarized target depends on the particle spin and can be written as

$$\hat{f}(E'_1, 0) = d + d_1 (\hat{S} \cdot \vec{e})^2, \quad (22)$$

where $\hat{S}$ is the deuteron spin operator and $\vec{e}$ is the unit vector along the deuteron momentum $\vec{k}$.

Substituting the expression for $\hat{f}(0)$ (22) in (21) one can obtain for a particle with spin $S = 1$:

$$\hat{V}_{\text{eff}} = -\frac{2\pi\hbar^2}{m_1\gamma_1\gamma_2} N_2 (d + d_1 (\hat{S} \cdot \vec{e})^2). \quad (23)$$

Let the quantization axis z is directed along $\vec{e}$ and M denotes the magnetic quantum number. Then, for a particle in the eigenstate of the operator of spin projection onto the z -axis $\hat{S}_z$, the effective potential energy can be written as

$$\hat{V}_{\text{eff}} = -\frac{2\pi\hbar^2}{m_1\gamma_1\gamma_2} N_2 (d + d_1 M^2). \quad (24)$$

According to (24), splitting of the deuteron energy levels in matter is similar to splitting of the atom energy levels in an electric field due to the quadratic Stark effect. Therefore, the above effect could be considered as caused by splitting of the particle spin levels in the pseudoelectric nuclear field of matter.

Expressions (23) and (8) are similar; therefore, the spin of a particle with $S \geq 1$ passing through nonpolarized matter rotates and oscillates (birefringence effect) [7, 8] due to interaction with the pseudoelectric nuclear field of matter. Henceforth, the notation $\hat{V}_{\text{E}}^{\text{nucl}} = \hat{V}_{\text{eff}}$ will be used for expression (23).

4. Spin rotation of a proton (deuteron, antiproton) in a storage ring with a polarized target

Let us now consider the experiments which deal with the use of polarized beams and targets. In this case the amplitude of the elastic coherent scattering at the zero angle depends on the vector polarization $\vec{P}_t$ of the target nuclei (if the spin of the target nuclei $J \geq 1$, then the addition depending on the target tensor polarization also appears; this addition will be omitted in following considerations, while the general case is considered in [4, 8]).

For the sake of concreteness suppose the target to be rest ($\gamma_2 = 1$, $\gamma_1 = \gamma$).

The contribution to the amplitude $\hat{f}(0)$ dependent on the vector polarization of the target nuclei can be expressed as

$$\hat{f}(\vec{P}_t, 0) = A_1(\hat{S} \cdot \vec{P}_t) + A_2(\hat{S} \cdot \vec{e})(\vec{e} \cdot \vec{P}_t), \quad (25)$$

where $\hat{S}$ is the operator of the particle spin. Contributions from weak P,T-odd interactions (see [4, 27–29]) are omitted here.

Correspondingly, the contribution to the effective potential energy of particle interaction with matter caused by the polarization of the target nuclei looks like

$$\hat{V}_{\text{eff}}(\vec{P}_t) = -\frac{2\pi\hbar^2}{m\gamma} N(A_1(\hat{S} \cdot \vec{P}_t) + A_2(\hat{S} \cdot \vec{e})(\vec{e} \cdot \vec{P}_t)). \quad (26)$$

Expression (26) can be rewritten as

$$\hat{V}_G^{\text{nucl}} \equiv \hat{V}_{\text{eff}}(\vec{P}_t) = -(\vec{\mu} \cdot \vec{G}) = -\frac{\mu}{S}(\hat{S} \cdot \vec{G}), \quad (27)$$

where μ is the particle magnetic moment,

$$\vec{G} = \frac{2\pi\hbar^2 S}{m\gamma\mu} N(A_1\vec{P}_t + A_2\vec{e}(\vec{e} \cdot \vec{P}_t)). \quad (28)$$

Recall now that the energy of the interaction of the magnetic moment $\vec{\mu}$ with a magnetic field $\vec{B}$ is as follows:

$$V_{\text{mag}} = -(\vec{\mu} \cdot \vec{B}) = -\frac{\mu}{S}(\hat{S} \cdot \vec{B}). \quad (29)$$

Expressions (27) and (29) are identical. Therefore, $\vec{G}$ can be interpreted as the effective pseudomagnetic field acting on the spin of the particle in matter with polarized nuclei and appears due to the nuclear interaction of the incident particles with scatterers. Similar to particle spin precession in an external magnetic field, spin also precesses in the field $\vec{G}$. This phenomenon was called the nuclear precession of the particle spin and described for the first time for slow neutrons in [11] and observed in [12–14].

Thus, when the polarized nuclear target is placed inside the storage ring, the spin of particles passing through the target additionally rotates under the action of the nuclear pseudomagnetic field. The question arises as to whether the pseudomagnetic nuclear field influences the particle momentum similarly to the Stern–Gerlach forces. It is known that the Stern–Gerlach forces are too weak to influence the motion of particles in the storage ring [19–23], i.e., they should not be included into the equations of particle motion. The pseudomagnetic nuclear field of the gas target is much lower than external magnetic fields in the storage ring; therefore, the pseudomagnetic analogue of the Stern–Gerlach forces also should not be included into the particle motion equations. Only the action of the pseudomagnetic field on spin motion is important.

5. The equations describing spin motion for a particle in a storage ring

5.1. Interactions contributing to the spin motion of a particle in a storage ring

According to the above analysis, when considering the motion of the spin of a particle in a storage ring, one should take into account several interactions:

- (1) interactions of the magnetic and electric dipole moments with electromagnetic fields;
- (2) interaction of the particle with electric fields due to the tensor electric polarizability;
- (3) interaction of the particle with magnetic fields due to the tensor magnetic polarizability;
- (4) interaction of the particle with the pseudoelectric nuclear field of matter;
- (5) interaction of the particle with the pseudomagnetic nuclear field of polarized matter.

The equation for the particle spin wavefunction including all these interactions is as follows:

$$i\hbar \frac{\partial \Psi(t)}{\partial t} = (\hat{H}_0 + \hat{V}_{\text{EDM}} + \hat{V}_{\vec{E}} + \hat{V}_{\vec{B}} + \hat{V}_{\text{E}}^{\text{nucl}} + \hat{V}_{\text{G}}^{\text{nucl}}) \Psi(t), \quad (30)$$

where $\Psi(t)$ is the particle spin wavefunction, $\hat{H}_0$ is the Hamiltonian describing spin motion caused by the interaction of the magnetic moment with the electromagnetic field (equation (30) with the only $\hat{H}_0$ summand converts to the Thomas–Bargmann–Michel–Telegdi equation); $\hat{V}_{\text{EDM}}$ describes interaction of the particle EDM d with the electric field; $\hat{V}_{\vec{E}}$ describes the interaction of the particle with the electric field due to the tensor electric polarizability; $\hat{V}_{\vec{B}}$ is the energy of interaction of the particle with the magnetic field due to the tensor magnetic polarizability; $\hat{V}_{\text{E}}^{\text{nucl}}$ describes the effective potential energy of particle interaction with the pseudoelectric field of the target (23). $\hat{V}_{\text{G}}^{\text{nucl}}$ describes the effective potential energy of particle interaction with the pseudomagnetic field of the target (27).

5.2. Deuteron spin rotation in electromagnetic fields of a storage ring

Let us consider a deuteron moving in a storage ring without targets inside and suppose the pressure of the residual gas to be low (10^{-10} Torr.). In this case the effects caused by the interactions $\hat{V}_{\text{E}}^{\text{nucl}}$ and $\hat{V}_{\text{G}}^{\text{nucl}}$ can be omitted. According to the above analysis, the motion of the deuteron spin, nevertheless, cannot be described by the Thomas–Bargmann–Michel–Telegdi equation: the interactions $\hat{V}_{\vec{E}}$ and $\hat{V}_{\vec{B}}$ should be considered. The corresponding equations can be written as follows [27–29]:

$$\begin{cases} \frac{d\vec{S}}{dt} = [\vec{S} \times \vec{\Omega}(\vec{d})] \\ -\frac{2}{3} \frac{\alpha_{\text{T}} E_{\text{eff}}^2}{\hbar} [\vec{e}_{\text{E}} \times \vec{e}_{\text{E}}'] - \frac{2}{3} \frac{\beta_{\text{T}} B_{\text{eff}}^2}{\hbar} [\vec{e}_{\text{B}} \times \vec{e}_{\text{B}}'], \\ \frac{dP_{ik}}{dt} = -(\varepsilon_{jkr} P_{ij} \Omega_r(d) + \varepsilon_{jir} P_{kj} \Omega_r(d)) \\ -\frac{3}{2} \frac{\alpha_{\text{T}} E_{\text{eff}}^2}{\hbar} ([\vec{e}_{\text{E}} \times \vec{S}]_i e_{\text{E},k} + e_{\text{E},i} [\vec{e}_{\text{E}} \times \vec{S}]_k) \\ -\frac{3}{2} \frac{\beta_{\text{T}} B_{\text{eff}}^2}{\hbar} ([\vec{e}_{\text{B}} \times \vec{S}]_i e_{\text{B},k} + e_{\text{B},i} [\vec{e}_{\text{B}} \times \vec{S}]_k), \end{cases} \quad (31)$$

where

$$\vec{\Omega}(d) = \vec{\Omega} + \vec{\Omega}_{\text{d}},$$

$$\vec{\Omega} = \frac{e}{mc} \left\{ \left(a + \frac{1}{\gamma} \right) \vec{B} - \left(\frac{g}{2} - \frac{\gamma}{\gamma+1} \right) \vec{\beta} \times \vec{E} \right\},$$

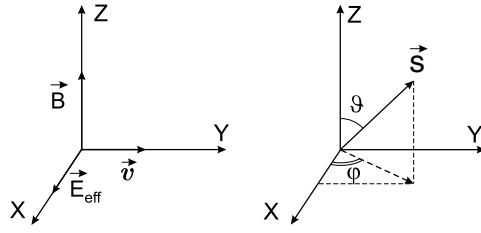


Figure 2. The coordinate system and vectors.

$$\vec{\Omega}_d = \frac{d}{\hbar} (\vec{E} + \vec{\beta} \times \vec{B}),$$

$$P_{ik} = \text{Tr } \rho \hat{Q}_{ik},$$

$$\hat{Q}_{ik} = \frac{3}{2} \left[S_i S_k + S_k S_i - \frac{4}{3} \delta_{ik} \right],$$

ρ is the spin density matrix of the particle, m is the particle mass, e is its charge, P_{ik} is the spin polarization tensor ($P_{xx} + P_{yy} + P_{zz} = 0$), γ is the Lorentz factor, $\vec{\beta} = \vec{v}/c$, $\vec{v}$ is the particle velocity, $a = (g - 2)/2$ (where g is the gyromagnetic ratio), $\vec{E}$ and $\vec{B}$ are the electric and magnetic fields at the point of particle location, $\vec{E}_{\text{eff}} = (\vec{E} + \vec{\beta} \times \vec{B})$, $\vec{B}_{\text{eff}} = (\vec{B} - \vec{\beta} \times \vec{E})$, $\vec{e}_E = \frac{\vec{E} + \vec{\beta} \times \vec{B}}{|\vec{E} + \vec{\beta} \times \vec{B}|}$, $\vec{e}_B = \frac{\vec{B} - \vec{\beta} \times \vec{E}}{|\vec{B} - \vec{\beta} \times \vec{E}|}$, $e'_{E,i} = P_{ik} e_{E,k}$, $e'_{B,i} = P_{il} e_{B,l}$ and $\Omega_r(d)$ are the components of the vector $\vec{\Omega}(d)$ ($r = 1, 2, 3$ correspond to x, y, z , respectively).

The equations for particle spin motion (31) can be rewritten as follows:

$$\begin{aligned} \frac{d\vec{S}}{dt} &= [\vec{S} \times \vec{\Omega}(d)] + \Omega_t [\vec{e}_E \times \vec{e}'_E] + \Omega_t^\mu [\vec{e}_B \times \vec{e}'_B], \\ \frac{dP_{ik}}{dt} &= -(\epsilon_{jkr} P_{ij} \Omega_r(d) + \epsilon_{jir} P_{kj} \Omega_r(d)) + \Omega'_t ([\vec{e}_E \times \vec{S}]_i e_{E,k} + e_{E,i} [\vec{e}_E \times \vec{S}]_k) \\ &\quad + \Omega_t'^\mu ([\vec{e}_B \times \vec{S}]_i e_{B,k} + e_{B,i} [\vec{e}_B \times \vec{S}]_k) \end{aligned} \quad (32)$$

where

$$\begin{aligned} \Omega_t &= -\frac{2}{3} \frac{\alpha_t E_{\text{eff}}^2}{\hbar}, & \Omega'_t &= -\frac{3}{2} \frac{\alpha_t E_{\text{eff}}^2}{\hbar}, & \Omega_t' &= -\frac{2}{3} \Omega_t, \\ \Omega_t^\mu &= -\frac{2}{3} \frac{\beta_t B_{\text{eff}}^2}{\hbar}, & \Omega_t'^\mu &= -\frac{3}{2} \frac{\beta_t B_{\text{eff}}^2}{\hbar}, & \Omega_t'^\mu &= -\frac{2}{3} \Omega_t^\mu. \end{aligned}$$

Suppose the external electric field in the storage ring $\vec{E} = 0$ and the particle moves along the circular orbit.

Let us now consider equation (32) in the coordinate system that rotates with the frequency of particle velocity rotation. In such a system the spin rotates with respect to the momentum with the frequency determined by $(g - 2)$. The coordinate system and vectors $\vec{v}$, $\vec{E}$, $\vec{B}$ are shown in figure 2. The axes x, y, z are denoted by the indices 1, 2, 3, respectively. As a result, the system (31) can be written as

$$\begin{aligned} \frac{dS_1}{dt} &= \Omega S_2 - \Omega_t^\mu P_{23}, \\ \frac{dS_2}{dt} &= -\Omega S_1 + (\Omega_t^\mu - \Omega_t) P_{13} + \omega_d S_3, \\ \frac{dS_3}{dt} &= \Omega_t P_{12} - \omega_d S_2, \end{aligned} \quad (33)$$

where $\omega_d = \frac{dE_l^{\text{eff}}}{\hbar}$,

$$\begin{aligned}\frac{dP_{11}}{dt} &= 2\Omega P_{12} + 2\omega_d P_{23}, \\ \frac{dP_{22}}{dt} &= -2\Omega P_{12}, \\ \frac{dP_{33}}{dt} &= -2\omega_d P_{23},\end{aligned}\tag{34}$$

$$\begin{aligned}\frac{dP_{12}}{dt} &= -\Omega (P_{11} - P_{22}) - \Omega'_t S_3 + \omega_d P_{13}, \\ \frac{dP_{13}}{dt} &= \Omega P_{23} + \Omega'_t S_2 - \Omega''_t S_2 - \omega_d P_{12}, \\ \frac{dP_{23}}{dt} &= -\Omega P_{13} + \Omega''_t S_1 - \omega_d (P_{22} - P_{33}).\end{aligned}\tag{35}$$

Remember that $P_{11} + P_{22} + P_{33} = 0$ and $P_{ik} = P_{ki}$.

5.3. Contribution from the EDM and tensor polarizabilities to deuteron spin oscillation

Let us consider the system (33)–(35) more attentively.

Suppose that a deuteron has neither EDM nor tensor polarizabilities: in this case the system (33)–(35) can be expressed as

$$\frac{dS_1}{dt} = \Omega S_2, \quad \frac{dS_2}{dt} = -\Omega S_1, \quad \frac{dS_3}{dt} = 0,\tag{36}$$

$$\frac{dP_{11}}{dt} = 2\Omega P_{12}, \quad \frac{dP_{22}}{dt} = -2\Omega P_{12}, \quad \frac{dP_{33}}{dt} = 0,\tag{37}$$

$$\frac{dP_{12}}{dt} = -\Omega (P_{11} - P_{22}), \quad \frac{dP_{13}}{dt} = \Omega P_{23}, \quad \frac{dP_{23}}{dt} = -\Omega P_{13}.\tag{38}$$

This is the conventional system of T-BMT equations that describes particle spin rotation with the frequency equal to $(g - 2)$ precession frequency $\Omega = \frac{ea}{mc} B$. The component S_3 of the vector polarization in these conditions is equal to a constant ($\frac{dS_3}{dt} = 0$), as well as the component P_{33} of the tensor polarization ($\frac{dP_{33}}{dt} = 0$).

Suppose the deuteron EDM differs from zero. Then the above system of equations converts to

$$\frac{dS_1}{dt} = \Omega S_2, \quad \frac{dS_2}{dt} = -\Omega S_1 + \omega_d S_3, \quad \frac{dS_3}{dt} = -\omega_d S_2,\tag{39}$$

$$\frac{dP_{11}}{dt} = 2\Omega P_{12} + 2\omega_d P_{23}, \quad \frac{dP_{22}}{dt} = -2\Omega P_{12}, \quad \frac{dP_{33}}{dt} = -2\omega_d P_{23},\tag{40}$$

$$\begin{aligned}\frac{dP_{12}}{dt} &= -\Omega (P_{11} - P_{22}) + \omega_d P_{13}, \\ \frac{dP_{13}}{dt} &= \Omega P_{23} - \omega_d P_{12}, \\ \frac{dP_{23}}{dt} &= -\Omega P_{13} - \omega_d (P_{22} - P_{33}).\end{aligned}\tag{41}$$

From (39)–(41) it follows that the presence of the nonzero EDM makes the vertical component S_3 of the vector polarization oscillating with the frequency of $(g - 2)$ precession Ω .

According to the idea [26], these oscillations can be eliminated by the modulation of the deuteron beam velocity with the frequency Ω :

$$v = v_0 + \delta v \sin(\Omega t + \varphi_f). \quad (42)$$

Here φ_f is the phase of the forced velocity oscillations.

As E_{eff} depends on $\vec{\beta} = \vec{v}/c$, it also appears modulated:

$$E_{\text{eff}} = E_{\text{eff}}^0 + \delta E_{\text{eff}} \sin(\Omega t + \varphi_f). \quad (43)$$

Therefore, $\omega_d = \frac{dE_{\text{eff}}}{\hbar}$ is also modulated with the same frequency. This makes the product $\omega_d S_2 \sim \sin^2(\Omega t + \varphi_f)$. From (39) it follows that

$$\begin{aligned} S_3(t) &= S_3(0) - \int_0^t \omega_d(t') S_2(t') dt' \\ &= S_3(0) + \text{const} \left[\frac{t}{2} - \frac{1}{4\Omega} (\sin(2\Omega t + 2\varphi_f)) - \sin 2\varphi_f \right]. \end{aligned}$$

According to (44), the component $S_3(t)$ contains both the linearly growing term and the term which oscillates with the frequency Ω and has a low amplitude. For better measurement conditions it is important to make $S_3(0) = 0$, i.e., the particle spin should initially lay in the horizontal plane. This is the reason to chose particle spin laying in the horizontal plane.

Similar reasoning shows that $P_{33}(t)$ also contains a linearly growing component. However, when the spin lays in the horizontal plane $P_{33}(0) \neq 0$.

It is important to note that when spin orientation meets the condition $\cos^2 \vartheta = \frac{1}{3}$ ($\cos \vartheta = \sqrt{\frac{1}{3}}$, $\sin \vartheta = \sqrt{\frac{2}{3}}$) one gets the component $P_{33}(0) = 0$, while $S_3(0) \neq 0$.

Therefore, taking ϑ that meets the condition $\cos \vartheta = \sqrt{\frac{1}{3}}$, one can use the component P_{33} for EDM measurements, too.

Let us consider now the contribution from the electric and magnetic tensor polarizabilities. Then, the system (33)–(35) should be considered instead of the system (39)–(41).

Some interesting implications follow from (33)–(35). As has already been mentioned above, in the experiments for the EDM search it is planned to measure the growth of the vertical component S_3 .

According to (33), the dependence of S_3 on time is described by the equation

$$\frac{dS_3}{dt} = \Omega_t P_{12} - \omega_d S_2 \quad (45)$$

and is determined by both the EDM and tensor polarizability of the deuteron. It is interesting that the derivative of the tensor polarization component P_{33} does not contain contributions from the tensor electric polarizability and is proportional only to the EDM:

$$\frac{dP_{33}}{dt} = -2\omega_d P_{23}. \quad (46)$$

Therefore, it is important to measure the component P_{33} , too. According to the above, spin orientation for this case is determined by the condition $\cos^2 \vartheta = \frac{1}{3}$.

5.4. Contribution from the tensor magnetic polarizability to deuteron spin oscillation

Contributions to spin rotation and oscillations from the EDM and polarizabilities are small. Therefore, they can be considered as perturbations to the complete system (33)–(35) and the role of each can be studied separately.

The system of equations taking into account the contribution from the tensor magnetic polarizability β_t is as follows:

$$\begin{aligned} \frac{dS_1}{dt} &= \Omega S_2 - \Omega_t^\mu P_{23}, & \frac{dS_2}{dt} &= -\Omega S_1 + \Omega_t^\mu P_{13}, \\ \frac{dP_{13}}{dt} &= \Omega P_{23} - \Omega_t'^\mu S_2, & \frac{dP_{23}}{dt} &= -\Omega P_{13} + \Omega_t'^\mu S_1. \end{aligned} \quad (47)$$

Introducing new variables $S_+ = S_1 + iS_2$ and $G_+ = P_{13} + iP_{23}$ and recomposing equations (47) to determine S_+ and G_+ , one obtains

$$\frac{dS_+}{dt} = -i\Omega S_+ + i\Omega_t^\mu G_+, \quad \frac{dG_+}{dt} = -i\Omega G_+ + i\Omega_t'^\mu S_+. \quad (48)$$

Let us search $S_+ \sim \tilde{S}_+ e^{-i\omega t}$, $G_+ \sim \tilde{G}_+ e^{-i\omega t}$, then (48) transforms as follows:

$$\omega \tilde{S}_+ = \Omega \tilde{S}_+ - \Omega_t^\mu \tilde{G}_+, \quad \omega \tilde{G}_+ = \Omega \tilde{G}_+ - \Omega_t'^\mu \tilde{S}_+. \quad (49)$$

The solution of this system can be easily found in the form

$$(\omega - \Omega)^2 - \Omega_t^\mu \Omega_t'^\mu = 0 \quad (50)$$

that finally gives

$$\omega_{1,2} = \Omega \pm \sqrt{\Omega_t^\mu \Omega_t'^\mu}. \quad (51)$$

The solution of (48) can be written as

$$S_+(t) = c_1 e^{-i\omega_1 t} + c_2 e^{-i\omega_2 t} = |c_1| e^{-i(\omega_1 t - \delta_1)} + |c_2| e^{-i(\omega_2 t - \delta_2)}. \quad (52)$$

Therefore,

$$S_1(t) = |c_1| \cos(\omega_1 t - \delta_1) + |c_2| \cos(\omega_2 t - \delta_2), \quad (53)$$

$$S_2(t) = -|c_1| \sin(\omega_1 t - \delta_1) - |c_2| \sin(\omega_2 t - \delta_2). \quad (54)$$

According to (53) and (54), the nonzero deuteron tensor magnetic polarizability makes the spin rotate with two frequencies ω_1 and ω_2 instead of Ω . The frequency difference $\Delta\omega = \omega_1 - \omega_2 = 2\sqrt{\Omega_t^\mu \Omega_t'^\mu} = \frac{\beta_t B_{\text{eff}}^2}{\hbar}$.

Let us recall now that EDM interaction with the electric field makes the deuteron spin rotate around the direction of this field and leads to the appearance of the S_3 component proportional to $S_2(t)$:

$$\frac{dS_3}{dt} \sim -\omega_d S_2. \quad (55)$$

Therefore,

$$\frac{dS_3}{dt} = \omega_d (|c_1| \sin(\omega_1 t - \delta_1) + |c_2| \sin(\omega_2 t - \delta_2)). \quad (56)$$

According to the idea [26], to measure EDM one should modulate the particle velocity ($v = v_0 + \delta v \sin(\Omega_f t + \varphi_f)$) with the frequency Ω_f that is close to the frequency Ω of $(g - 2)$ precession.

If we suppose the magnetic polarizability to be equal to zero, then $\omega_1 = \omega_2 = \Omega$ and the spin rotates in the horizontal plane with frequency Ω . In this case velocity modulation with the same frequency $\Omega_f = \Omega$ gives

$$\frac{dS_3}{dt} \sim \sin^2(\Omega t) \quad (57)$$

and the vertical component can be written as $S_3 = \int_0^t \frac{dS_3}{dt'} dt' \sim \int_0^t \sin^2 \Omega t' dt' \sim \frac{1}{2}(t - \frac{1}{2\Omega} \sin \Omega t)$. The vertical component apparently includes the term, which linearly grows with time, and the low amplitude term oscillating with high frequency.

However, $\omega_1 \neq \omega_2$ and velocity modulation, for example, with the frequency $\Omega = \omega_1$ provides for slow spin oscillation with the frequency $(\omega_1 - \omega_2)$ instead of linear growth. According to the evaluation [34], the tensor magnetic polarizability $\beta_t \sim 2 \times 10^{-40}$; therefore, in the field $B \sim 10^4$ gauss the beatings frequency $\Delta\omega \sim 10^{-5}$. Measurement of the frequency of the beatings makes possible to find the tensor magnetic polarizability of the deuteron (or a nucleus).

Thus, due to the presence of the tensor magnetic polarizability, the horizontal component of the spin rotates around $\vec{B}$ with two frequencies ω_1, ω_2 instead of expected rotation with the frequency Ω . This is the reason for the component S_3 to experience oscillations, similar to those caused by the EDM. Therefore, particle velocity modulation with the frequency Ω provides for eliminating oscillations with this frequency, but S_3 oscillations with the frequency $\Delta\omega$ rest (similarly P_{33}). A study of these oscillations is necessary, because they can distort the EDM measurements.

5.5. Contribution from the tensor electric polarizability to deuteron spin oscillation

Let us now consider the contribution caused by the tensor electric polarizability. From the system (34) it follows that

$$\begin{aligned} \frac{d(P_{11} - P_{22})}{dt} &= 4\Omega P_{12}, \\ \frac{d^2 P_{12}}{dt^2} &= -\Omega \frac{d(P_{11} - P_{22})}{dt} - \Omega'_t \frac{dS_3}{dt} = -(4\Omega^2 + \Omega_T \Omega'_t) P_{12}. \end{aligned} \quad (58)$$

Thus, we have the equation

$$\frac{d^2 P_{12}}{dt^2} + \omega_{12}^2 P_{12} = 0, \quad (59)$$

where $\omega_{12} = \sqrt{4\Omega^2 + \Omega_T \Omega'_t} \approx 2\Omega$, because $\Omega_T \Omega'_t \ll \Omega^2$.

The solution of this equation can be found in the form

$$P_{12} = c_1 \cos \omega_{12} t + c_2 \sin \omega_{12} t. \quad (60)$$

Let us find the coefficients c_1 and c_2 : when $t = 0$ equation (60) gives $c_1 = P_{12}(0)$. The coefficient c_2 can be found from

$$\left. \frac{dP_{12}}{dt} \right|_{(t \rightarrow 0)} = \omega_{12} c_2. \quad (61)$$

Therefore,

$$c_2 = \frac{1}{\omega_{12}} \left. \frac{dP_{12}}{dt} \right|_{(t \rightarrow 0)}. \quad (62)$$

From equation (35)

$$\left. \frac{dP_{12}}{dt} \right|_{(t \rightarrow 0)} = -\Omega (P_{11}|_{(t \rightarrow 0)} - P_{22}|_{(t \rightarrow 0)}), \quad (63)$$

it follows that

$$c_2 = -\frac{P_{11} - P_{22}}{2} \quad (64)$$

and

$$P_{12} = P_{12}(0) \cos \omega_{12}t - \frac{P_{11} - P_{22}}{2} \sin \omega_{12}t. \quad (65)$$

As a result, one can write the following equation for the vertical component of the spin S_3 :

$$\frac{dP_3}{dt} = \Omega_t P_{12}(t) = \Omega_t \left[P_{12}(0) \cos 2\Omega t - \frac{P_{11}(0) - P_{22}(0)}{2} \sin 2\Omega t \right]. \quad (66)$$

As can be seen, the vertical component of the spin oscillates with frequency 2Ω .

The equation describing contribution to S_3 from the tensor electric polarizability and EDM looks like (33):

$$\frac{dS_3}{dt} = \Omega_t P_{12} - \omega_d S_2. \quad (67)$$

As the frequency of S_2 oscillations is Ω , the product $\omega_d S_2$ contains non-oscillating terms (see section 5.3). As a result, the contribution to S_3 , which is caused by the EDM, either linearly grows with time, when $\Omega_f = \Omega$, or slowly oscillates with the frequency $\Omega_f - \Omega$, when $\Omega_f \neq \Omega$.

It is important that modulation of the velocity $v = v_0 + \delta v \sin(\Omega_f t + \varphi_f)$ results in E_{eff} oscillation (see (43)) and, therefore, E_{eff}^2 also oscillates with time and appears proportional to $\sin^2(\Omega_f t + \varphi_f)$. As a result, the contribution to S_3 caused by the tensor electric polarizability can be expressed as

$$\frac{dS_3(\alpha_t)}{dt} \sim \Delta\Omega_t \sin^2(\Omega_f t + \varphi_f) \left[P_{12}(0) \cos 2\Omega t - \frac{P_{11}(0) - P_{22}(0)}{2} \sin 2\Omega t \right], \quad (68)$$

where $\Delta\Omega_t = -\frac{2}{3} \frac{\alpha_t}{\hbar} (\delta E_{\text{eff}})^2$, i.e.,

$$\frac{dS_3(\alpha_t)}{dt} \sim -\frac{1}{2} \Delta\Omega_t \cos(2\Omega_f t + 2\varphi_f) \left[P_{12}(0) \cos 2\Omega t - \frac{P_{11}(0) - P_{22}(0)}{2} \sin 2\Omega t \right]. \quad (69)$$

According to (69), for a partially polarized deuteron beam the derivative $\frac{dS_3}{dt}$ depends on the deuteron polarization components P_{12} and $\frac{P_{11}(0) - P_{22}(0)}{2}$.

For simplicity let us consider the deuteron beam in the pure polarization state. In this case the components P_{12} and $\frac{P_{11}(0) - P_{22}(0)}{2}$ can be calculated using the explicit expression for the spin wavefunctions as follows. Suppose $\vec{e}(\vartheta, \varphi)$ is the unit vector directed along the deuteron spin (ϑ and φ are the polar and azimuth angles (see figure 2)). So the spin wavefunction that describes the deuteron spin state with the magnetic quantum number $m = 1$ can be expressed as follows (in the Cartesian basis):

$$\chi_1(\vartheta, \varphi) = \begin{pmatrix} a_x \\ a_y \\ a_z \end{pmatrix} = -\frac{1}{\sqrt{2}} \begin{pmatrix} \cos \vartheta \cos \varphi - i \sin \varphi \\ \cos \vartheta \sin \varphi + i \cos \varphi \\ -\sin \vartheta \end{pmatrix}. \quad (70)$$

The polarization vector can be written as

$$\vec{S} = \langle \hat{S} \rangle = \chi_1^\dagger \hat{S} \chi_1 = i[\vec{a} \times \vec{a}^*] \quad (71)$$

and the components of the polarization tensor

$$\langle P_{ik} \rangle = \chi_1^\dagger \hat{Q}_{ik} \chi_1 = -\frac{3}{2} \{ a_i a_k^* + a_k a_i^* - \frac{2}{3} \}, \quad (72)$$

where $\hat{Q}_{ik}$ is the spin tensor of rank 2 ($\hat{Q}_{ik} = \frac{1}{2}(\hat{S}_i \hat{S}_k + \hat{S}_k \hat{S}_i - \frac{4}{3} \delta_{ik} S)$). Therefore,

$$P_{12} = \frac{3}{4} \sin 2\varphi \sin^2 \vartheta, \quad (73)$$

$$\frac{P_{11} - P_{22}}{2} = \frac{3}{4} \cos 2\varphi \sin^2 \vartheta, \quad (74)$$

$$P_{33} = -\frac{3}{2} \left(\sin^2 \vartheta - \frac{2}{3} \right). \quad (75)$$

Using (49), (53), (54) one can obtain

$$\frac{dS_3(\alpha_t)}{dt} \sim -\frac{3}{8} \Delta\Omega_t \sin^2 \vartheta \cos(2\Omega_t t + 2\varphi_f) [\sin 2\varphi \cos 2\Omega t - \cos 2\varphi \sin 2\Omega t]. \quad (76)$$

From (76) it follows that $\frac{dS_3}{dt}$ slowly oscillates with the frequency $(\Omega_f - \Omega)$.

In the ideal case, when $\Omega_f = \Omega$ (as it is proposed in [26] for EDM measurement) (76) converts to

$$\frac{dS_3(\alpha_t)}{dt} = -\frac{3}{8} \Delta\Omega_t \sin^2 \vartheta \cos(2\Omega t + 2\varphi_f) \sin(2\Omega t - 2\varphi). \quad (77)$$

In the general case, when the phases φ_f and φ are arbitrary, (77) contains terms that do not depend on time and, therefore, S_3 linearly grows with time, as the signal from the EDM does.

It is interesting that choosing $\varphi_f = -\varphi$ one obtains

$$\frac{dS_3(\alpha_t)}{dt} \sim \cos(2\Omega t - 2\varphi) \sin(2\Omega t - 2\varphi) = \frac{1}{2} \sin(4\Omega t - 4\varphi) \quad (78)$$

which makes this contribution to S_3 quickly oscillating and depressed. But even in this ideal case (in real situation $\Omega \neq \Omega_f$, though) it rests the contribution caused by the tensor magnetic polarizability (9).

The measurement of these contribution allows us to find the tensor electric polarizability. According to evaluations [34], $\alpha_t \sim 10^{-40} \text{ cm}^3$, therefore, for the field $E_{\text{eff}} = B \sim 10^4$ gauss the frequency $\Omega_t \sim 10^{-5} \text{ s}^{-1}$. When considering modulation, one can estimate $\Delta\Omega_t \sim \Omega_t \left(\frac{\delta v}{v_0} \right)^2$, then, suppose $\left(\frac{\delta v}{v_0} \right)^2 \sim 10^{-2}-10^{-3}$, it can be obtained $\Delta\Omega_t \sim 10^{-7}-10^{-8} \text{ s}^{-1}$ that exceeds the magnitude of ω_d for the deuteron EDM $d = 10^{-29} \text{ e cm}$.

6. Particle spin rotation in a storage ring with an internal target

Let now a gas target be placed inside the storage ring. According to section 5.1, spin motion in this case is also influenced by the nuclear pseudoelectric and pseudomagnetic fields.

The description of spin motion appears complicated, because multiple scattering, which occurs in the target, brings changes in both the angular spread and depolarization of the particle beam and should be taken into account.

So, the spin density matrix formalism should be used to derive equations for the evolution of the particle spin.

The density matrix of the system ‘deuteron+target’ is

$$\rho = \rho_p \otimes \rho_t, \quad (79)$$

where ρ_p is the density matrix of the particle beam and ρ_t is the density matrix of the target.

The equation for the particle density matrix can be written as [6, 27]

$$\frac{d\rho_p}{dt} = -\frac{i}{\hbar} [\hat{H}_p, \rho_p] + \left(\frac{\partial \rho_p}{\partial t} \right)_{\text{col}}, \quad (80)$$

where $\hat{H}_p$ is the Hamiltonian describing particle motion in external electromagnetic fields and $\left(\frac{\partial \rho_p}{\partial t} \right)_{\text{col}}$ describes the evolution of the density matrix due to collisions with the target atoms (nuclei).

The collision term $\left(\frac{\partial \rho_p}{\partial t}\right)_{\text{col}}$ can be expressed as follows:

$$\left(\frac{\partial \rho_p}{\partial t}\right)_{\text{col}} = vN \text{Tr}_t \left[\frac{2\pi i}{k} [F_{(\theta=0)} \rho - \rho F_{(\theta=0)}^+] \right] \quad (81)$$

$$+ \int d\Omega F(\vec{k}') \rho(\vec{k}') F^+(\vec{k}'), \quad (82)$$

where $\vec{k}' = \vec{k} + \vec{q}$, the vector $\vec{q}$ is the momentum transferred to a nucleus of matter from the incident particle, v is the speed of the incident particles, N is the atom density in matter, F is the scattering amplitude depending on the spin operators of the deuteron and the matter nucleus (atom), F^+ is the Hermitian conjugate for the operator F . The first term in (81) describes coherent elastic scattering of the particle by matter nuclei, while the second term is for multiple scattering. Let us consider the first term in (81). It can be written as

$$\left(\frac{\partial \rho_p}{\partial t}\right)_{\text{col}}^{(1)} = vN \frac{2\pi i}{k} [\hat{f}(0) \rho_p - \rho_p \hat{f}(0)^+]. \quad (83)$$

The amplitude $\hat{f}(0)$ of particle scattering in the target at the zero angle is

$$\hat{f}(0) = \text{Tr}_t \hat{F}(0) \rho_t. \quad (84)$$

As a result, one can obtain

$$\left(\frac{\partial \rho_p}{\partial t}\right)_{\text{col}}^{(1)} = -\frac{i}{\hbar} (\hat{V}_{\text{eff}} \rho_p - \rho_p \hat{V}_{\text{eff}}^+), \quad (85)$$

where

$$\begin{aligned} \hat{V}_{\text{eff}} &= -\frac{2\pi\hbar^2}{m\gamma} N \hat{f}(0) \\ &= -\frac{2\pi\hbar^2}{m\gamma} N (d + A_1(\hat{S} \cdot \vec{P}_t) + A_2(\hat{S} \cdot \vec{e})(\vec{e} \cdot \vec{P}_t) + d_1(\hat{S} \cdot \vec{e})^2). \end{aligned} \quad (86)$$

The latter expression can be rewritten as

$$\hat{V}_{\text{eff}} = -\frac{2\pi\hbar^2}{m\gamma} Nd - \frac{\mu}{S} (\vec{S} \cdot \vec{G}) - \frac{2\pi\hbar^2}{m\gamma} Nd_1(\hat{S} \cdot \vec{e})^2, \quad (87)$$

where $\vec{G} = \frac{2\pi\hbar^2 S}{m\gamma\mu} N(A_1 \vec{P}_t + A_2 \vec{e}(\vec{e} \cdot \vec{P}_t))$; see (28). For particles with the spin $\frac{1}{2}$ the term proportional to d_1 is absent.

Finally, expression (80) reads

$$\frac{d\rho_p}{dt} = -\frac{i}{\hbar} [\hat{H}_p, \rho_p] - \frac{i}{\hbar} (\hat{V}_{\text{eff}} \rho_p - \rho_p \hat{V}_{\text{eff}}^+) + vN \text{Tr}_t \int d\Omega F(\vec{k}') \rho(\vec{k}') F^+(\vec{k}'). \quad (88)$$

The last term in the above formula, which is proportional to Tr_t , describes the multiple scattering process and spin depolarization aroused from it.

When studying the interaction of particles with the target inside the storage ring, the density of the target is chosen to make multiple scattering in it small for the observation time; the depolarization of the beam also appears small.

For high-energy particles scattering at small angles is important. Therefore, most of the scattered particles remain on their orbits. For this case the latter term should be considered.

Further analysis of equations (88) in the present paper will be done for such time of the experiment that makes possible to cast out the last term in (88).

As a result, we have

$$\frac{d\rho_p}{dt} = -\frac{i}{\hbar} [\hat{H}_p, \rho_p] - \frac{i}{\hbar} (\hat{V}_{\text{eff}} \rho_p - \rho_p \hat{V}_{\text{eff}}^+). \quad (89)$$

Now the equation which describes the evolution of the spin properties of a particle in a storage ring can be obtained. The polarization vector is as follows:

$$\vec{P} = \frac{\text{Tr} \rho_p(t) \hat{S}}{\text{Tr} \rho_p(t)} = \frac{\text{Tr} \rho_p(t) \hat{S}}{I(t) S}, \quad (90)$$

where $I(t) = \text{Tr} \rho_p(t)$ is the beam intensity.

From (90) one can obtain the differential equation allowing us to find beam polarization:

$$\frac{d\vec{P}}{dt} = \frac{\text{Tr} \frac{d\rho_p(t)}{dt} \hat{S}}{SI(t)} - \vec{P} \frac{1}{I(t)} \frac{dI(t)}{dt} \quad (91)$$

When the particle spin is $S = 1$, the polarization tensor should also be found:

$$P_{ik} = \frac{\text{Tr} \rho_p \hat{Q}_{ik}}{I(t)}. \quad (92)$$

The change in the tensor polarization with time can be written as

$$\frac{dP_{ik}}{dt} = \frac{1}{I(t)} \text{Tr} \left(\frac{d\rho_p}{dt} \hat{Q}_{ik} \right) - P_{ik} \frac{1}{I(t)} \frac{dI(t)}{dt}, \quad (93)$$

$\hat{Q}_{ik} = \frac{3}{2}(S_i S_k + S_k S_i - \frac{4}{3} \delta_{ik})$. Using the equation for the density matrix (88) and expressions (91), (93) one can obtain the equation for the case where spin motion is influenced by the pseudoelectric and pseudomagnetic nuclear fields.

Let us first consider particles with spin $S = \frac{1}{2}$. The density matrix for such a particle can be expressed as follows:

$$\rho_{p\frac{1}{2}} = I_{\frac{1}{2}}(t) \left(\frac{1}{2} \hat{I} + (\vec{P} \cdot \hat{S}) \right), \quad (94)$$

where $\hat{I}$ is the unit matrix in the spin space and $\hat{S}$ is the spin operator.

With the help of (90) it is possible to find that

$$\frac{dI_{\frac{1}{2}}(t)}{dt} = -\frac{i}{\hbar} \text{Tr} (\hat{V}_{\text{eff}} \rho_{p\frac{1}{2}} - \rho_{p\frac{1}{2}} \hat{V}_{\text{eff}}^+) = -\left(\kappa + \frac{2\mu}{\hbar} (\vec{G}_2 \cdot \vec{P}) \right) I_{\frac{1}{2}}(t), \quad (95)$$

where $\kappa = vN\sigma_{\text{tot}}$, v is the particle speed, σ_{tot} is the total cross-section of particle scattering by a nonpolarized nucleus and $\vec{G}_2$ is the imaginary part of the pseudomagnetic nuclear field. Expression (95) can be rewritten as

$$\frac{dI_{\frac{1}{2}}(t)}{dt} = -\kappa_{\text{abs}}(\vec{P}) I_{\frac{1}{2}}(t), \quad (96)$$

where the absorption coefficient $\kappa_{\text{abs}} = vN[\sigma_{\text{tot}} + \frac{1}{2}\sigma_1(\vec{P} \cdot \vec{P}_t) + \frac{1}{2}\sigma_2(\vec{P} \cdot \vec{e})(\vec{e} \cdot \vec{P}_t)] = \kappa + \vec{P} \cdot \vec{g}_t$ and $\vec{g}_t = \frac{1}{2}vN(\sigma_1 \vec{P}_t + \sigma_2 \vec{e})(\vec{e} \cdot \vec{P}_t)$.

The cross-section $\sigma_1 = \sigma_{\text{tot}}^{\uparrow\uparrow}(\vec{e} \perp \vec{P}_t) - \sigma_{\text{tot}}^{\uparrow\downarrow}(\vec{e} \perp \vec{P}_t)$ is the difference between the cross-sections of particle scattering by a polarized nucleus with $\vec{P}$ parallel ($\vec{P} \uparrow\uparrow \vec{P}_t$) and antiparallel ($\vec{P} \uparrow\downarrow \vec{P}_t$) to the target polarization vector $\vec{P}_t$ in conditions when the particle momentum is orthogonal to the target polarization vector ($\vec{e} \perp \vec{P}_t$). The cross-section $\sigma_2 = \sigma_{\text{tot}}^{\uparrow\uparrow}(\vec{e} \parallel \vec{P}_t) - \sigma_{\text{tot}}^{\uparrow\downarrow}(\vec{e} \parallel \vec{P}_t)$ is the difference between the cross-sections of particle scattering by a polarized nucleus with the vector $\vec{P}$ parallel ($\vec{P} \uparrow\uparrow \vec{P}_t$) and antiparallel ($\vec{P} \uparrow\downarrow \vec{P}_t$) to the target polarization vector $\vec{P}_t$ in conditions when the particle momentum is parallel to the target polarization vector ($\vec{e} \parallel \vec{P}_t$).

Equation (96) describes the well-known fact that the coefficient of particle beam absorption in a polarized target depends on the respective orientations of the beam and target polarization vectors.

Let us obtain now the equation that describes the evolution of the particle vector polarization under the action of the pseudomagnetic nuclear field $\vec{G}$.

According to (90), (91) one can obtain

$$\begin{aligned} \frac{d\vec{P}}{dt} &= \frac{1}{SI_{\frac{1}{2}}(t)} \text{Tr} \frac{d\rho_p}{dt} \hat{S} - \vec{P} \frac{1}{I_{\frac{1}{2}}(t)} \frac{dI_{\frac{1}{2}}(t)}{dt} \\ &= -\frac{2\mu}{\hbar} [\vec{G}_1 \times \vec{P}] - \frac{2\mu}{\hbar} (\vec{G}_2 - \vec{P}(\vec{G}_2 \cdot \vec{P})), \end{aligned} \quad (97)$$

where $\vec{G}_1 = \text{Re } \vec{G}$, $\vec{G}_2 = \text{Im } \vec{G}$.

Equation (97) can be rewritten as

$$\frac{d\vec{P}}{dt} = [\vec{P} \times \vec{\Omega}_{\text{nuc}}] - (\vec{g}_t - \vec{P}(\vec{P} \cdot \vec{g}_t)), \quad (98)$$

where $\vec{\Omega}_{\text{nuc}}$ is the frequency of spin precession in the pseudomagnetic nuclear field

$$\vec{\Omega}_{\text{nuc}} = \frac{2\mu}{\hbar} \vec{G}_1 = \frac{2\pi\hbar}{m\gamma} N(\text{Re } A_1 \vec{P}_t + \text{Re } A_2 \vec{e}(\vec{e} \cdot \vec{P}_t)). \quad (99)$$

Adding the contribution from the pseudomagnetic nuclear field (98) to the T-BMT equation (32), one can obtain the equation describing spin evolution for a particle moving in a storage ring with a polarized target inside:

$$\frac{d\vec{P}}{dt} = [\vec{P} \times (\vec{\Omega}(d) + \vec{\Omega}_{\text{nuc}})] - (\vec{g}_t - \vec{P}(\vec{P} \cdot \vec{g}_t)), \quad (100)$$

$$\frac{dI_{\frac{1}{2}}(t)}{dt} = -(\chi + (\vec{P} \cdot \vec{g}_t))I_{\frac{1}{2}}(t), \quad (101)$$

where $\vec{g}_t = \frac{1}{2}vN(\sigma_1 \vec{P}_t + \sigma_2 \vec{e}(\vec{e} \cdot \vec{P}_t))$.

Let us remind here that equation (100), similar to the T-BMT equation, describes the polarization vector of a particle moving along a certain trajectory. Strictly speaking, to find the polarization vector of a particle beam moving in a storage ring $\vec{P}$ should be averaged over the particle distribution in the beam phase space [19–24].

The contribution from the pseudomagnetic nuclear field to the spin motion of the $S = 1$ particle (for example, deuteron) can be obtained in a similar way. The spin density matrix for the spin 1 particle is expressed as follows:

$$\rho_1 = I_1(t) \left(\frac{1}{3} \hat{I} + \frac{1}{2} (\vec{P} \cdot \hat{S}) + \frac{1}{9} P_{ik} \hat{Q}_{ik} \right). \quad (102)$$

The effective potential energy of deuteron interaction with a polarized target is expressed as $\hat{V}_{\text{eff}} = -\frac{2\pi\hbar^2}{m\gamma} Nd - \frac{\mu}{S} (\hat{S} \cdot \vec{G}) - \frac{2\pi\hbar^2}{m\gamma} Nd_1 (\hat{S} \cdot \vec{e})^2$ (see (87)).

The energy $\hat{V}_{\text{eff}}$ for a deuteron (in contrast to a particle with the spin $S = \frac{1}{2}$) contains both deuteron interaction with the pseudomagnetic field and pseudoelectric field (the term proportional to d_1). The equations describing deuteron spin rotation (birefringence effect) in the pseudoelectric field of the nonpolarized internal target in the storage ring were obtained in [27].

If the internal target in the storage ring is polarized, then using (87), (89) and (102) one can find the change in the beam intensity and polarization vector as follows:

$$\frac{dI_1(t)}{dt} = - \left[(\chi + 2(\vec{g}_t \cdot \vec{P})) - \frac{\chi}{3} (2 + P_{ik} e_i e_k) \right] I_1(t), \quad (103)$$

where $\chi = -\frac{4\pi v N}{k} \text{Im} d_1 = -vN(\sigma_{M=1} - \sigma_{M=0})$, $\chi = vN\sigma_{M=0}$, $\sigma_{M=1}$ and $\sigma_{M=0}$ are the total cross-sections of deuteron scattering by a nonpolarized nucleus for the deuteron state with the

magnetic quantum number $M = 1$ and $M = 0$, respectively (the quantization axis is directed along $\vec{e}$).

The polarization vector can be expressed as

$$\begin{aligned} \frac{d\vec{P}}{dt} = & [\vec{P} \times \vec{\Omega}_{\text{nuc}}] - \frac{2}{3} P_{ik} g_{tk} - \frac{4}{3} \vec{g}_t + 2\vec{P}(\vec{P} \cdot \vec{g}_t) + \frac{\eta}{3} [\vec{e} \times \vec{e}'] \\ & + \frac{\chi}{2} (\vec{e}(\vec{e} \cdot \vec{P}) + \vec{P}) - \frac{2\chi}{3} \vec{P} - \frac{\chi}{3} (\vec{e} \cdot \vec{e}') \vec{P}, \end{aligned} \quad (104)$$

where $\eta = -\frac{4\pi N}{k} \text{Re } d_1$ and $e'_i = P_{ik} e_k$.

According to (104), the nuclear pseudomagnetic field causes deuteron spin precession with the frequency $\vec{\Omega}_{\text{nuc}}$. Contribution (104) should be added to equation (32) to make them describe motion of the deuteron spin in the storage ring with the target inside.

A similar approach allows us to describe the evolution of tensor polarization (93), but it is too bulky to adduce it here.

However, when the particle beam absorption in the target can be neglected, one can make all changes in the equation, which describes deuteron spin behavior, by replacement of $\vec{\Omega}(d)$ with the new value $(\vec{\Omega}(d) + \vec{\Omega}_{\text{nuc}})$.

Note that as the target inside the storage ring is of the finite size, the particle moving in the storage ring interacts with the target at times (the expressions for $\vec{\Omega}_{\text{nuc}}$ and $\vec{g}_t$ contain the density N , which means the density at the point of particle location i.e. $N = N(t)$). This is the reason for $\vec{\Omega}_{\text{nuc}}$ and $\vec{g}_t$ to oscillate with the frequency of particle rotation in the storage ring Ω_0 .

For further analysis let us expand $\vec{\Omega}_{\text{nuc}}$ and $\vec{g}_t$ into Fourier series and leave only zero harmonic. In this case $\vec{\Omega}_{\text{nuc}}$ and $\vec{g}_t$ appear to be constants and can be written as

$$\begin{aligned} \vec{\Omega}_{\text{nuc}} = & \frac{2\pi\hbar}{m\gamma} N (\text{Re } A_1 \vec{P}_t + \text{Re } A_2 \vec{e}(\vec{e} \cdot \vec{P}_t)) \frac{l}{vT} \\ = & \frac{2\pi\hbar}{m\gamma} \frac{j_t}{L} (\text{Re } A_1 \vec{P}_t + \text{Re } A_2 \vec{e}(\vec{e} \cdot \vec{P}_t)) \\ = & \frac{2\pi\hbar}{m\gamma} \frac{j_t v}{v} (\text{Re } A_1 \vec{P}_t + \text{Re } A_2 \vec{e}(\vec{e} \cdot \vec{P}_t)), \end{aligned} \quad (105)$$

where l is the target length, $j_t = N \cdot l$ is the target density in cm^{-2} (usually for the polarized gas target $j_t \approx 10^{14} \text{ cm}^{-2}$), T is the period of beam rotation in the storage ring, v is the frequency of particle rotation in the storage ring, L is the particle orbit length inside the storage ring.

The zero harmonic for $\vec{g}_t$ is as follows:

$$\begin{aligned} \vec{g}_t = & \frac{1}{2} v N (\sigma_1 \vec{P}_t + \sigma_2 \vec{e}(\vec{e} \cdot \vec{P}_t)) \frac{l}{vT} \\ = & \frac{1}{2} j_t v (\sigma_1 \vec{P}_t + \sigma_2 \vec{e}(\vec{e} \cdot \vec{P}_t)). \end{aligned} \quad (106)$$

Let us evaluate now the effect magnitude. For protons, antiprotons and deuterons with energy from MeV to GeV the value $\text{Re } A_1 \sim 10^{-12} - 10^{-13} \text{ cm}$. Considering $v \sim 10^6 \text{ s}^{-1}$ one gets for the nuclear precession frequency $\Omega_{\text{nuc}} = 10^{-4} - 10^{-5} \text{ s}^{-1}$.

Note (see (18) and (19)) that the ratio $\frac{1}{\gamma} \text{Re } A_{1(2)}$ is proportional to the T-matrix and, as it follows, Ω_{nuc} depends on the particle energy only due to the possible dependence of the T-matrix on energy.

The obtained estimation for Ω_{nuc} allows us to expect, for example for COSY, to measure the spin rotation angle $\vartheta = \Omega_{\text{nuc}} t \approx 10^{-2} - 10^{-3} \text{ rad}$ in the observation time $t \sim 10^2 \text{ s}$. This value is quite observable.

To prevent the suppression of the spin precession in the pseudomagnetic nuclear field by the storage ring magnetic field $\vec{B}$, one can use either pseudomagnetic nuclear field which rotates with the $(g - 2)$ precession frequency [29] or other methods for $(g - 2)$ precession cancellation (see, for example, [17, 25, 26]).

Let us now consider the particular example. Suppose the axis OZ is orthogonal to the orbit plane ($OZ \parallel \vec{B}$) (see figure 2). The pseudomagnetic field is directed along the axis OX . In this case for the particle in the straight section of the storage ring the vertical spin component rotates around the direction of the pseudomagnetic field in the ZOY plane. The change of the vertical (or horizontal) spin component should be observed.

As the typical spin rotation angle $\vartheta = \Omega_{\text{nuc}} t \ll 1$, the dependence of the vertical spin component on time can be expressed as

$$S_3(t) = S_3(0) - \frac{1}{2}\vartheta^2 t^2 = S_3(0) - \Omega_{\text{nuc}}^2 t^2. \quad (107)$$

The effect can be strengthened when adding a magnetic field $\vec{b} \gg \vec{G}_1$ directed along the pseudomagnetic field $\vec{G}_1$ (along the axis OX in the case under consideration). In this case, the angle of rotation is expressed as $\vartheta = \vartheta_{\text{mag}} + \vartheta_{\text{nuc}} = (\Omega_{\text{mag}}(\vec{b}) + \Omega_{\text{nuc}}) \cdot t$ and the vertical component $S_3(t) = S_3(0) - \frac{1}{2}\vartheta_{\text{mag}}^2 t^2 - \vartheta_{\text{mag}}\vartheta_{\text{nuc}}$ (the terms $\sim \vartheta_{\text{nuc}}^2$ are neglected).

When making the direction of the field $\vec{b}$ change during the experiment (i.e. the sign of ϑ_{mag} changing), one gets the possibility of measuring the difference $S_3(\vec{b} \uparrow \vec{G}_1, t) - S_3(\vec{b} \downarrow \vec{G}_1, t) = 2\vartheta_{\text{mag}}\vartheta_{\text{nuc}}$. This permits us to study the possibility of effect observation and, therefore, to measure the spin-dependent part of the amplitude of elastic coherent zero-angle scattering.

7. Conclusion

In the present paper equations describing the motion of the spin of a particle in a storage ring are obtained by considering contributions from the tensor electric and magnetic polarizabilities of the particle along with contributions from the spin rotation and birefringence effects in the polarized matter of the gas target.

The influence of the tensor electric and magnetic polarizabilities on spin motion in the resonance deuteron EDM experiment is considered in detail.

It is shown that, besides EDM, the electric and magnetic polarizabilities also contribute to the vertical spin component S_3 . Moreover, the electric polarizability contributes to the S_3 component, even when the deuteron EDM is supposed to be zero and, thereby, the electric polarizability can imitate the EDM contribution. It is shown that, unlike the vertical component of the spin S_3 , the component P_{33} of the polarization tensor does not contain contribution from the electric polarizability, whereas contribution from the magnetic polarizability reveals only when the deuteron EDM differs from zero.

It is also shown that when the angle ϑ between the spin direction and the vertical axis meets the condition $\sin \vartheta = \sqrt{\frac{2}{3}}$ ($\cos \vartheta = \sqrt{\frac{1}{3}}$), the initial value of P_{33} appears $P_{33}(0) = 0$. As a result, contribution from EDM to the measured signal linearly grows with time starting from zero. Therefore, the measurement of the P_{33} component of the deuteron tensor polarization seems to be of particular interest, especially because the nonzero component P_{33} unambiguously indicates the EDM presence (in contrast to the S_3 component, whose appearance can be aroused by the tensor electric polarizability rather than EDM).

It is also shown that the study of the spin rotation and birefringence effects for a high-energy particle in a storage ring provides for the measurement of the spin-dependent real part of the coherent elastic forward scattering amplitude.

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